

1. Administrative & Study Identification

Full Study Title: Phase 1/2 Study of Intravesical MK-3120 in BCG-Naïve or BCG-Exposed High-Risk Non-muscle Invasive Bladder Cancer
Short Title/Acronym: A Clinical Study of MK-3120 in People With Bladder Cancer (MK-3120-003) **Protocol Number:** MK-3120-003 **NCT Number:** NCT07222488 **Sponsor Name:** Merck Sharp & Dohme LLC
Investigational Product: MK-3120 (Biological) **Phase of Development:** Phase 1/Phase 2 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of MK-3120 when administered intravesically.
Secondary Objectives: To determine if the use of MK-3120 is effective in treating high-risk non-muscle invasive bladder cancer (HR NMIBC) after transurethral resection of the bladder tumor (TURBT).

2.2 Background & Rationale Researchers are looking for new ways to treat high-risk non-muscle invasive bladder cancer (HR NMIBC). NMIBC is cancer in the tissue that lines the inside of the bladder and has not spread to the bladder muscle or outside of the bladder. In standard treatment for HR NMIBC, doctors first remove the tumor with a procedure called transurethral resection of the bladder tumor (TURBT). This study aims to evaluate if intravesical administration of the study medicine, MK-3120, is a safe and tolerable therapeutic option post-TURBT, accommodating both BCG-naïve and BCG-exposed patient populations.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 45 evaluable participants **Number of Sites:** Multiple global locations (e.g., USA, Canada) **Estimated Study Start:** 09-Dec-2025 **Estimated Primary Completion:** 28-Feb-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years. 2. Histologically confirmed carcinoma in situ (CIS) +/- papillary high-risk non-

muscle invasive bladder cancer (NMIBC), confirmed locally. 3. Individual whose most recent transurethral resection of bladder tumor (TURBT) was performed within 12 weeks before allocation and showed high-risk NMIBC histology (for papillary tumors Ta and T1, a visually complete resection must have been performed). 4. Is either Bacillus Calmette-Guérin (BCG)-naïve or BCG-exposed (with recurrence $\geq$ 12 months but $\leq$ 24 months after the last BCG dose).

4.2 Exclusion Criteria 1. History of or current locally advanced (i.e., T2, T3, T4) or metastatic urothelial cancer (UC). 2. Concurrent extravesical (i.e., urethra, ureter, renal pelvis) non-muscle invasive UC or history of extravesical non-muscle invasive UC that recurred within the last 2 years. 3. Active total bladder incontinence, active urinary tract infection, neurogenic bladder, urethral stricture, or any condition that would prohibit normal voiding (or holding bladder voiding for 1 to 2 hours). 4. Known active central nervous system metastases, carcinomatous meningitis, or active infection requiring systemic therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: MK-3120 administered once weekly for the first 6 weeks, followed by once monthly for 9 months. **Route of Administration:** Intravesical **Duration of Treatment:** Up to ~10.5 months of exposure.

5.2 Concomitant & Prohibited Therapy Permitted: Stable antiretroviral therapy for HIV-positive participants; antiviral therapy for HBV/HCV (providing undetectable viral loads). **Prohibited:** Systemic therapy for active infections; current immunosuppressive steroids for pneumonitis/ILD.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-84 to -70	Screening
2	0 to 14	Baseline
3	14 to 28	Interim
4	28 to 42	Interim
5	42 to 56	Interim
6	56 to 70	Interim
7	70 to 84	Interim
8	84 to 98	Interim
9	98 to 112	Interim
10	112 to 126	Interim
11	126 to 140	Interim
12	140 to 154	Interim
13	154 to 168	Interim
14	168 to 182	Interim
15	182 to 196	Interim
16	196 to 210	Interim
17	210 to 224	Interim
18	224 to 238	Interim
19	238 to 252	Interim
20	252 to 266	Interim
21	266 to 280	Interim
22	280 to 294	Interim
23	294 to 308	Interim
24	308 to 322	Interim
25	322 to 336	Interim
26	336 to 350	Interim
27	350 to 364	Interim
28	364 to 378	Interim
29	378 to 392	Interim
30	392 to 406	Interim
31	406 to 420	Interim
32	420 to 434	Interim
33	434 to 448	Interim
34	448 to 462	Interim
35	462 to 476	Interim
36	476 to 490	Interim
37	490 to 504	Interim
38	504 to 518	Interim
39	518 to 532	Interim
40	532 to 546	Interim
41	546 to 560	Interim
42	560 to 574	Interim
43	574 to 588	Interim
44	588 to 602	Interim
45	602 to 616	Interim
46	616 to 630	Interim
47	630 to 644	Interim
48	644 to 658	Interim
49	658 to 672	Interim
50	672 to 686	Interim
51	686 to 700	Interim
52	700 to 714	Interim
53	714 to 728	Interim
54	728 to 742	Interim
55	742 to 756	Interim
56	756 to 770	Interim
57	770 to 784	Interim
58	784 to 798	Interim
59	798 to 812	Interim
60	812 to 826	Interim
61	826 to 840	Interim
62	840 to 854	Interim
63	854 to 868	Interim
64	868 to 882	Interim
65	882 to 896	Interim
66	896 to 910	Interim
67	910 to 924	Interim
68	924 to 938	Interim
69	938 to 952	Interim
70	952 to 966	Interim
71	966 to 980	Interim
72	980 to 994	Interim
73	994 to 1008	Interim
74	1008 to 1022	Interim
75	1022 to 1036	Interim
76	1036 to 1050	Interim
77	1050 to 1064	Interim
78	1064 to 1078	Interim
79	1078 to 1092	Interim
80	1092 to 1106	Interim
81	1106 to 1120	Interim
82	1120 to 1134	Interim
83	1134 to 1148	Interim
84	1148 to 1162	Interim
85	1162 to 1176	Interim
86	1176 to 1190	Interim
87	1190 to 1204	Interim
88	1204 to 1218	Interim
89	1218 to 1232	Interim
90	1232 to 1246	Interim
91	1246 to 1260	Interim
92	1260 to 1274	Interim
93	1274 to 1288	Interim
94	1288 to 1302	Interim
95	1302 to 1316	Interim
96	1316 to 1330	Interim
97	1330 to 1344	Interim
98	1344 to 1358	Interim
99	1358 to 1372	Interim
100	1372 to 1386	Interim
101	1386 to 1400	Interim
102	1400 to 1414	Interim
103	1414 to 1428	Interim
104	1428 to 1442	Interim
105	1442 to 1456	Interim
106	1456 to 1470	Interim
107	1470 to 1484	Interim
108	1484 to 1498	Interim
109	1498 to 1512	Interim
110	1512 to 1526	Interim
111	1526 to 1540	Interim
112	1540 to 1554	Interim
113	1554 to 1568	Interim
114	1568 to 1582	Interim
115	1582 to 1596	Interim
116	1596 to 1610	Interim
117	1610 to 1624	Interim
118	1624 to 1638	Interim
119	1638 to 1652	Interim
120	1652 to 1666	Interim
121	1666 to 1680	Interim
122	1680 to 1694	Interim
123	1694 to 1708	Interim
124	1708 to 1722	Interim
125	1722 to 1736	Interim
126	1736 to 1750	Interim
127	1750 to 1764	Interim
128	1764 to 1778	Interim
129	1778 to 1792	Interim
130	1792 to 1806	Interim
131	1806 to 1820	Interim
132	1820 to 1834	Interim
133	1834 to 1848	Interim
134	1848 to 1862	Interim
135	1862 to 1876	Interim
136	1876 to 1890	Interim
137	1890 to 1904	Interim
138	1904 to 1918	Interim
139	1918 to 1932	Interim
140	1932 to 1946	Interim
141	1946 to 1960	Interim
142	1960 to 1974	Interim
143	1974 to 1988	Interim
144	1988 to 2002	Interim
145	2002 to 2016	Interim
146	2016 to 2030	Interim
147	2030 to 2044	Interim
148	2044 to 2058	Interim
149	2058 to 2072	Interim
150	2072 to 2086	Interim
151	2086 to 2100	Interim
152	2100 to 2114	Interim
153	2114 to 2128	Interim
154	2128 to 2142	Interim
155	2142 to 2156	Interim
156	2156 to 2170	Interim
157	2170 to 2184	Interim
158	2184 to 2198	Interim
159	2198 to 2212	Interim
160	2212 to 2226	Interim
161	2226 to 2240	Interim
162	2240 to 2254	Interim
163	2254 to 2268	Interim
164	2268 to 2282	Interim
165	2282 to 2296	Interim
166	2296 to 2310	Interim
167	2310 to 2324	Interim
168	2324 to 2338	Interim
169	2338 to 2352	Interim
170	2352 to 2366	Interim
171	2366 to 2380	Interim
172	2380 to 2394	Interim
173	2394 to 2408	Interim
174	2408 to 2422	Interim
175	2422 to 2436	Interim
176	2436 to 2450	Interim
177	2450 to 2464	Interim
178	2464 to 2478	Interim
179	2478 to 2492	Interim
180	2492 to 2506	Interim
181	2506 to 2520	Interim
182	2520 to 2534	Interim
183	2534 to 2548	Interim
184	2548 to 2562	Interim
185	2562 to 2576	Interim
186	2576 to 2590	Interim
187	2590 to 2604	Interim
188	2604 to 2618	Interim
189	2618 to 2632	Interim
190	2632 to 2646	Interim
191	2646 to 2660	Interim
192	2660 to 2674	Interim
193	2674 to 2688	Interim
194	2688 to 2702	Interim
195	2702 to 2716	Interim
196	2716 to 2730	Interim
197	2730 to 2744	Interim
198	2744 to 2758	Interim
199	2758 to 2772	Interim
200	2772 to 2786	Interim
201	2786 to 2800	Interim
202	2800 to 2814	Interim
203	2814 to 2828	Interim
204	2828 to 2842	Interim
205	2842 to 2856	Interim
206	2856 to 2870	Interim
207	2870 to 2884	Interim
208	2884 to 2898	Interim
209	2898 to 2912	Interim
210	2912 to 2926	Interim
211	2926 to 2940	Interim
212	2940 to 2954	Interim
213	2954 to 2968	Interim
214	2968 to 2982	Interim
215	2982 to 2996	Interim
216	2996 to 3010	Interim
217	3010 to 3024	Interim
218	3024 to 3038	Interim
219	3038 to 3052	Interim
220	3052 to 3066	Interim
221	3066 to 3080	Interim
222	3080 to 3094	Interim
223	3094 to 3108	Interim
224	3108 to 3122	Interim
225	3122 to 3136	Interim
226	3136 to 3150	Interim
227	3150 to 3164	Interim
228	3164 to 3178	Interim
229	3178 to 3192	Interim
230	3192 to 3206	Interim
231	3206 to 3220	Interim
232	3220 to 3234	Interim
233	3234 to 3248	Interim
234	3248 to 3262	Interim
235	3262 to 3276	Interim
236	3276 to 3290	Interim
237	3290 to 3304	Interim
238	3304 to 3318	Interim
239	3318 to 3332	Interim
240	3332 to 3346	Interim
241	3346 to 3360	Interim
242	3360 to 3374	Interim
243	3374 to 3388	Interim
244	3388 to 3402	Interim
245	3402 to 3416	Interim
246	3416 to 3430	Interim
247	3430 to 3444	Interim
248	3444 to 3458	Interim
249	3458 to 3472	Interim
250	3472 to 3486	Interim
251	3486 to 3500	Interim
252	3500 to 3514	Interim
253	3514 to 3528	Interim
254	3528 to 3542	Interim
255	3542 to 3556	Interim
256	3556 to 3570	Interim
257	3570 to 3584	Interim
258	3584 to 3598	Interim
259	3598 to 3612	Interim
260	3612 to 3626	Interim
261	3626 to 3640	Interim
262	3640 to 3654	Interim
263	3654 to 3668	Interim
264	3668 to 3682	Interim
265	3682 to 3696	Interim
266	3696 to 3710	Interim
267	3710 to 3724	Interim
268	3724 to 3738	Interim
269	3738 to 3752	Interim
270	3752 to 3766	Interim
271	3766 to 3780	Interim
272	3780 to 3794	Interim
273	3794 to 3808	Interim
274	3808 to 3822	Interim
275	3822 to 3836	Interim
276	3836 to 3850	Interim
277	3850 to 3864	Interim
278	3864 to 3878	Interim
279	3878 to 3892	Interim
280	3892 to 3906	Interim
281	3906 to 3920	Interim
282	3920 to 3934	Interim
283	3934 to 3948	Interim
284	3948 to 3962	Interim
285	3962 to 3976	Interim
286	3976 to 3990	Interim
287	3990 to 4004	Interim
288	4004 to 4018	Interim
289	4018 to 4032	Interim
290	4032 to 4046	Interim
291	4046 to 4060	Interim
292	4060 to 4074	Interim
293	4074 to 4088	Interim
294	4088 to 4102	Interim
295	4102 to 4116	Interim
296	4116 to 4130	Interim
297	4130 to 4144	Interim
298	4144 to 4158	Interim
299	4158 to 4172	Interim
300	4172 to 4186	Interim
301	4186 to 4200	Interim
302	4200 to 4214	Interim
303	4214 to 4228	Interim
304	4228 to 4242	Interim
305	4242 to 4256	Interim
306	4256 to 4270	Interim
307	4270 to 4284	Interim
308	4284 to 4298	Interim
309	4298 to 4312	Interim
310	4312 to 4326	Interim
311	4326 to 4340	Interim
312	4340 to 4354	Interim
313	4354 to 4368	Interim
314	4368 to 4382	Interim
315	4382 to 4396	Interim
316	4396 to 4410	Interim
317	4410 to 4424	Interim
318	4424 to 4438	Interim
319	4438 to 4452	Interim
320	4452 to 4466	Interim
321	4466 to 4480	Interim
322	4480 to 4494	Interim
323	4494 to 4508	Interim
324	4508 to 4522	Interim
325	4522 to 4536	Interim
326	4536 to 4550	Interim
327	4550 to 4564	Interim
328	4564 to 4578	Interim
329	4578 to 4592	Interim
330	4592 to 4606	Interim
331	4606 to 4620	Interim
332	4620 to 4634	Interim
333	4634 to 4648	Interim
334	4648 to 4662	Interim
335	4662 to 4676	Interim
336	4676 to 4690	Interim
337	4690 to 4704	Interim
338	4704 to 4718	Interim
339	4718 to 4732	Interim
340	4732 to 4746	Interim
341	4746 to 4760	Interim
342	4760 to 4774	Interim
343	4774 to 4788	Interim
344	4788 to 4802	Interim
345	4802 to 4816	Interim
346	4816 to 4830	Interim
347	4830 to 4844	Interim
348	4844 to 4858	Interim
349	4858 to 4872	Interim
350	4872 to 4886	Interim
351	4886 to 4900	

Limiting Toxicities [DLTs]) **Measurement Method:** Clinical assessment and laboratory monitoring throughout the study.

7.2 Secondary & Exploratory Endpoints Evaluation of clinical efficacy, exploring complete response (CR) rates and disease recurrence rates in treating HR NMIBC after TURBT.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of treatment-emergent adverse events (TEAEs) throughout the trial, specifically focusing on tolerability of intravesical administration. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal/External safety monitoring mechanisms managed by Merck Sharp & Dohme LLC.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive at least one dose of MK-3120) for primary endpoint; Intent-to-Treat (ITT) population for exploratory efficacy outcomes. **Sample Size Justification:** Target enrollment of $N=45$ is considered sufficient for initial safety, tolerability, and preliminary efficacy signals in a Phase 1/2 setting. **Handling of Missing Data:** Standard Phase 1/2 imputation rules applied for safety parameters; Last Observation Carried Forward (LOCF) where applicable for efficacy.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote clinical monitoring visits to ensure subject safety and protocol compliance. **Audit Trail:** eCRF locking and data clarification forms via standard quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: MK-3120-003 **Registry Number:** NCT07222488 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Clinical Study of MK-3120 in People With Bladder Cancer (MK-3120-003) **Official Regulatory Title:** Phase 1/2 Study of Intravesical MK-3120 in BCG-Naïve or BCG-Exposed High-Risk Non-muscle Invasive Bladder Cancer

12. Clinical Framework (Hyperkalemia Specific - Adjusted for Bladder Cancer)

Primary Condition: High-Risk Non-muscle Invasive Bladder Cancer (HR NMIBC) **Diagnosis Threshold:** Histologically confirmed CIS +/- papillary high-risk NMIBC within 12 weeks of TURBT **Medical Methodology:** Surgical resection (TURBT) followed by intravesical pharmacological intervention **Optimization Target:** Disease eradication, prevention of recurrence, and maintenance of normal voiding function

13. Study Procedures & Methodology

Management Pathway: Standard TURBT followed by study-specific intravesical MK-3120 installation **Education Protocol (HCP):** Administration protocol training for intravesical biologic delivery **Education Protocol (Patient):** Counseling on expected urinary side effects, fluid intake restrictions around installation, and treatment scheduling **Quality Audit Mechanism:** Assessment of complete visually confirmed tumor resections prior to dosing and verification of appropriate inclusion histology

14. Observation & Follow-Up Schedule

Total Duration: Primary completion spanning approximately 3 years (Dec 2025 to Feb 2029) **Onsite Visit Cadence:** Weekly for the first 6 weeks, then every 4 weeks (monthly) for 9 months **Remote Monitoring Frequency:** Standard clinical trial frequency ensuring patient reporting of AEs. **Checklist Review Interval:** Routine follow-up per protocol specified intervals for oncological monitoring.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				